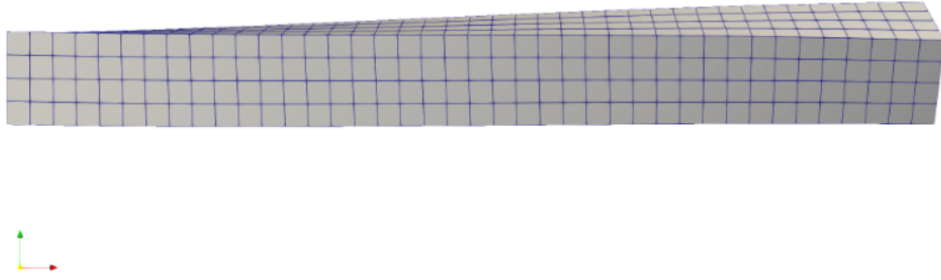


Weak forms and problem setup — Twisting of a Neo-Hookean Block

1 Governing equations

1.1 Geometry and boundary



The reference configuration is

$$\Omega = (0, L_x) \times (0, L_y) \times (0, L_z) \subset \mathbb{R}^3, \quad L_x = 10, \quad L_y = L_z = 1.$$

Partition the boundary:

$$\Gamma_L = \{x = 0\}, \quad \Gamma_R = \{x = L_x\}, \quad \Gamma_N = \partial\Omega \setminus (\Gamma_L \cup \Gamma_R).$$

1.2 Material model

Let $\mathbf{u} : \Omega \rightarrow \mathbb{R}^3$ be the displacement. Define

$$\mathbf{F} = \mathbf{I} + \nabla \mathbf{u}, \quad J = \det \mathbf{F}, \quad \mathbf{F}^{-T} = (\mathbf{F}^{-1})^T.$$

For Lamé parameters $\mu, \lambda > 0$, the first Piola–Kirchhoff stress is

$$\mathbf{P}(\mathbf{F}) = \mu \mathbf{F} + (\lambda \ln J - \mu) \mathbf{F}^{-T}.$$

1.3 Strong form

Seek $\mathbf{u}$ such that

$$\begin{aligned} -\nabla \cdot \mathbf{P}(\mathbf{u}) &= 0 && \text{in } \Omega, \\ \mathbf{u} &= \mathbf{0} && \text{on } \Gamma_L, \\ \mathbf{u} &= \mathbf{g} && \text{on } \Gamma_R, \\ \mathbf{P}(\mathbf{u}) \mathbf{n} &= 0 && \text{on } \Gamma_N, \end{aligned}$$

with prescribed displacement on Γ_R :

$$\mathbf{g}(x, y, z) = \begin{pmatrix} -0.1 \\ y(\cos \pi - 1) - z \sin \pi \\ y \sin \pi + z(\cos \pi - 1) \end{pmatrix}.$$

2 Weak form and derivations

Multiply the strong form by a test function $\mathbf{v} \in V$ and integrate over Ω :

$$0 = \int_{\Omega} (-\nabla \cdot \mathbf{P}) \cdot \mathbf{v} \, dx = - \int_{\partial\Omega} (\mathbf{P} \mathbf{n}) \cdot \mathbf{v} \, ds + \int_{\Omega} \mathbf{P} : \nabla \mathbf{v} \, dx.$$

On Γ_N we have $\mathbf{P} \mathbf{n} = 0$, and on $\Gamma_L \cup \Gamma_R$ the variation $\mathbf{v} = 0$, so the boundary integral vanishes. Hence the weak form:

$$\boxed{\int_{\Omega} \left[\mu \operatorname{tr}(\mathbf{F}^T \nabla \mathbf{v}) + (\lambda \ln J - \mu) \operatorname{tr}(\mathbf{F}^{-T} \nabla \mathbf{v}) \right] dx = 0 \quad \forall \mathbf{v} \in V.}$$

Derivation of the weak form

Starting from $0 = \int_{\Omega} (-\nabla \cdot \mathbf{P}) \cdot \mathbf{v} \, dx$, apply Gauss' theorem and note vanishing boundary terms as above. Rewriting $\mathbf{P} : \nabla \mathbf{v} = \operatorname{tr}(\mathbf{P}^T \nabla \mathbf{v})$ and substituting the expression for $\mathbf{P}(\mathbf{F})$ yields the boxed form.

2.1 Jacobian (Fréchet derivative)

Define the residual functional

$$R(\mathbf{u})[\mathbf{v}] = \int_{\Omega} \mathbf{P}(\mathbf{F}(\mathbf{u})) : \nabla \mathbf{v} \, dx.$$

Its Fréchet derivative in direction $\delta \mathbf{u}$ is

$$DR(\mathbf{u})[\delta \mathbf{u}, \mathbf{v}] = \left. \frac{d}{d\varepsilon} R(\mathbf{u} + \varepsilon \delta \mathbf{u})[\mathbf{v}] \right|_{\varepsilon=0} = \int_{\Omega} (\delta \mathbf{P}) : \nabla \mathbf{v} \, dx.$$

We compute variations:

$$\delta \mathbf{F} = \nabla \delta \mathbf{u}, \quad \delta J = J \operatorname{tr}(\mathbf{F}^{-1} \nabla \delta \mathbf{u}), \quad \delta \ln J = \operatorname{tr}(\mathbf{F}^{-1} \nabla \delta \mathbf{u}), \quad \delta(\mathbf{F}^{-T}) = -\mathbf{F}^{-T} (\nabla \delta \mathbf{u}) \mathbf{F}^{-T}.$$

Thus

$$\begin{aligned} \delta \mathbf{P} &= \mu \delta \mathbf{F} + (\lambda \ln J - \mu) \delta(\mathbf{F}^{-T}) + \lambda (\delta \ln J) \mathbf{F}^{-T} \\ &= \mu \nabla \delta \mathbf{u} - (\lambda \ln J - \mu) \mathbf{F}^{-T} (\nabla \delta \mathbf{u}) \mathbf{F}^{-T} + \lambda \operatorname{tr}(\mathbf{F}^{-1} \nabla \delta \mathbf{u}) \mathbf{F}^{-T}. \end{aligned}$$

Substitution gives the Jacobian bilinear form:

$$\boxed{DR(\mathbf{u})[\delta \mathbf{u}, \mathbf{v}] = \int_{\Omega} \left[\mu \operatorname{tr}((\nabla \delta \mathbf{u})^T \nabla \mathbf{v}) - (\lambda \ln J - \mu) \operatorname{tr}((\nabla \delta \mathbf{u} \mathbf{F}^{-1})^T (\nabla \mathbf{v} \mathbf{F}^{-1})) + \lambda \operatorname{tr}(\mathbf{F}^{-1} \nabla \delta \mathbf{u}) \operatorname{tr}(\mathbf{F}^{-1} \nabla \mathbf{v}) \right] dx.}$$